

Catalonia

Catalan Competition Authority

The Catalan Competition Authority (ACCO) launched the ERICCA project in 2022 with two main objectives: assisting the organisation's technicians in handling sanctioning files related to Catalan public tenders and using artificial intelligence (AI) to proactively identify public tenders with a higher probability of anti-competitive agreements among participating companies.

One of the initial main challenges of the project was obtaining quality data from Catalan public tenders. This problem was partially mitigated by signing a collaboration agreement with the Department of Economy of the Generalitat de Catalunya.² Thanks to this agreement, ACCO can directly access the database used by the Public Contracts Registry. Additionally, data from Dades Obertes is also utilised.³

The tool applies an artificial intelligence algorithm to the available data (an unsupervised machine learning model) to classify companies into clusters, taking into account, among other variables, the tenders they have participated in. Subsequently, various statistical parameters are calculated to identify clusters where collusion among the participating companies is more likely. Since mid-2023, the model has been complemented by a neural network.

Between 2023 and early 2024, the project has progressed in five key areas.

I. Expansion of Analysed Data

In previous versions of ERICCA, the tool did not analyse low-value tenders due to processing capacity limitations.⁴ However, in the latest version, thanks to improvements in computational efficiency, the tool also analyses "minor contracts."⁵ This enhancement allows the detection of behaviours prohibited by law, such as the artificial splitting of contracts.

² Initially, the information was obtained using web scrapping techniques applied to the Public Procurement Services Platform. Contractacio Publica, <https://contractaciopublica.cat/en/inici> (last accessed June 7, 2024).

³ Dades Obertes is a portal of the Generalitat de Catalunya (Government of Catalonia) that collects various datasets produced or compiled by public agencies that public administrations make available to citizens so that they can use them freely in a simple and convenient manner.

⁴ Under €15,000.

⁵ For amounts greater than €3,000.

II. Facilitation of File Processing

To improve the processing of files, new reports and visualisations have been implemented, offering a clearer and more detailed view of the analysed tenders. These functionalities include:

- **Global Visualisation of Tenders by Contracting Body:** This function shows all tenders issued by a specific body, including the awarded amounts and the average number of competitors by product or service type. This information is crucial for assessing the level of competition and detecting possible irregularities.
- **Scatter Plots to Identify Suspicious Behaviours:** When a contracting body and a product or service type are selected, the tool generates a scatter plot of the bids submitted by all participating companies, represented in different colours. This visualisation allows for the visual identification of behaviours that might indicate collusion among some companies.
- **Analysis of Company Groups:** This functionality shows all possible groups of companies that have participated in tenders, indicating the number of tenders each group has participated in and the awarded amount. Additionally, the user can select the number of companies forming the group (up to a maximum of eight) and view all possible combinations of the companies in the group.
- **Data Export in CSV Format:** To facilitate external analysis and integration with other data processing tools, ERICCA allows downloading all information in CSV format. This function enables technicians to write more detailed reports and use other data processing programmes, such as spreadsheets, for different types of analyses. These advances significantly ease the work of technicians in processing sanctioning files, providing more powerful and intuitive tools to detect and analyse anti-competitive behaviours.

III. New Statistical Parameters

New statistical parameters have been incorporated to provide more robustness to the tool's results. For example, the tool calculates the expected discount of a tender considering the contracting body, the number of competitors, or the type of product or service tendered and compares it with the obtained discount to identify possible anti-competitive behaviours. The 'random forest' method is also used to flag contracts with unusually low discounts.

IV. Use of a Neural Network

The proactive detection of possible collusion cases has been complemented with the use of a neural network based on the model developed by researchers David Imhof and Martin Huber.⁶ This neural network has learned from data on public tenders from Switzerland, Italy, and Japan, characterising price distributions in competitive and collusive scenarios to predict possible collusion in a given tender with a high level of confidence.⁷ The model calculates the bid interaction matrices for pairs of all companies given a set of tenders.

In the example shown below, the variability of the values in each region of this interaction matrix according to the competitive situation is displayed. Visually, clear differences between the two types of behaviours are already observed, but the algorithm can detect these and other more subtle differences to make predictions.

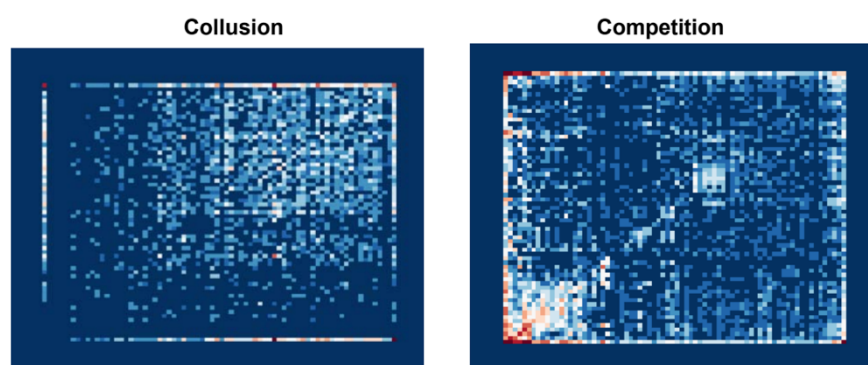


Figure I: Collusion and competition

V. Collaboration Agreements with Other Organisations and Entities

Given the potential of this tool, various institutions have contacted ACCO to use ERICCA. This tool can be used for functions not strictly related to detecting anti-competitive behaviours. As previously mentioned, the tool allows detecting contract splitting, which could violate public procurement regulations. In this regard, the possibility of providing access to some of ERICCA's functions and analyses to other institutions and public bodies is currently being evaluated.

⁶ Martin Huber & David Imhof, *Flagging cartel participants with deep learning based on convolutional neural networks* 89 INT. J. INDUS. ORG. 102946 (2023).

⁷ In future versions, the model will also learn from Catalan tender data.

The evolution of ERICCA represents a significant step forward in using artificial intelligence to detect and prevent anti-competitive practices in public tenders. This initiative not only improves the efficiency of tender oversight but also reinforces ACCO's commitment to transparency and fair competition in the public market.